LOCALIZATION OF CABLE FAULTS

TYPES OF CABLE FAULTS

The faults which are mostly likely to occur in the cables are:

1. Ground or Earth Faults

When the insulation of the cable gets damaged, the current starts flowing from core to earth or to cable sheath. Such faults are known as ground or earth faults.

2. Cross or Short-circuit faults

When the insulation between two cables or between two cores of a multi-core cable gets damaged, the current starts flowing from one cable to another cable or from one core to another core of a multicore cable directly (without passing through the load). Such faults are known as short-circuit faults.

3. Open-circuit faults

When the conductor of a cable is broken or joint is pulled out there is no current in the cable. Such faults are known as open-circuit faults.

First the nature of fault is determined and then the point of fault is located. For determination of nature of faults, the insulation resistance of each of each core to ground and between cores is measured with the aid of a megger. The low value of insulation resistance between any core and earth indicates the ground fault whereas the low value of insulation resistance between two cores (with far ends of the cable isolated from load) indicates short-circuit fault.

For determination of open-circuit fault, the far end of the cable, in case of single cable, is earthed (or the far ends of the cables are interconnected), an ammeter is inserted in series with the cable and low voltage supply is connected across the cable and earth or cables. Zero deflection in ammeter indicates open-circuit fault.

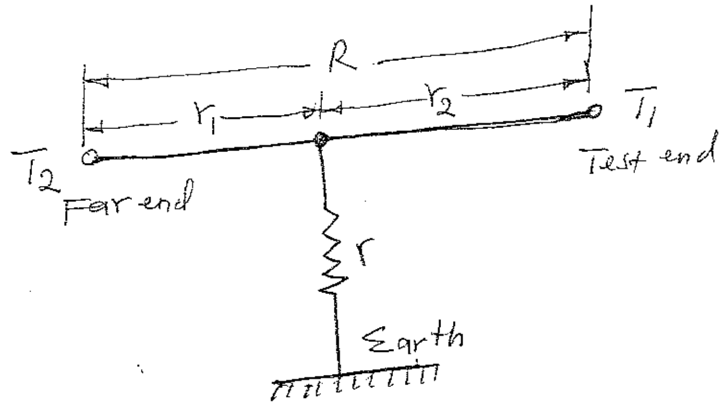
LOCATION OF CABLE FAULTS

These are various methods for locating the point of faults.

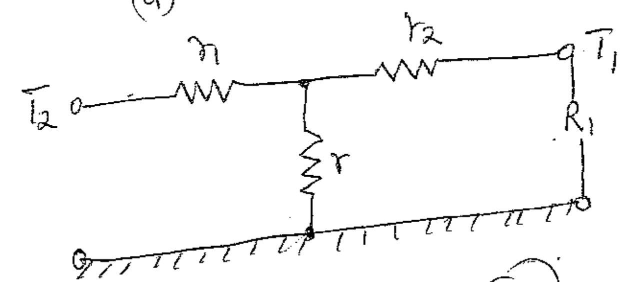
1. BLAVIER'S TEST

Blavier's test is used to locate the ground fault of a single cable i.e. when no other cables run along with the faulty one.

This test is performed with the aid of a low voltage supply and either an ammeter or voltmeter meter or a bridge network.



(a)



(b)

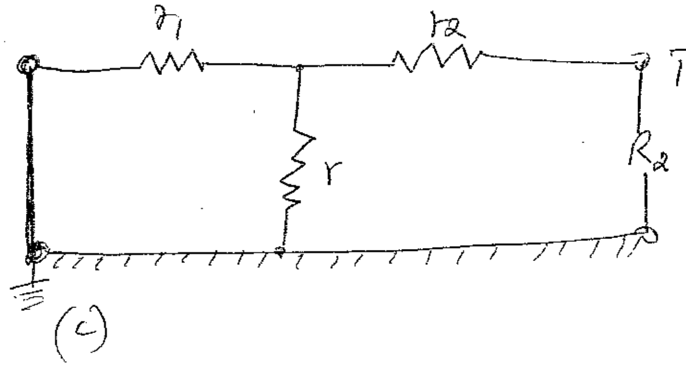


Figure 1: Blavier's test for Ground Fault of a single core cable

In this test resistance between one end of the cable T_1 and earth is measured first with the far end T_2 isolated from earth as in Figure 1(b) and then with the far end earthed as shown in Figure 1 (c). Let the two readings be R_1 and R_2 respectively.

If r_1 and r_2 are the conductor resistances of the lengths of cable 'far end' to fault and 'test end' to fault respectively and r is the resistance of fault to earth then

$$R_1 = r_2 + r \qquad \ldots (1)$$

and
$$R_2 = \frac{r_1 r}{r_1 + r} + r_2 \qquad \ldots (2)$$

The total resistance of the conductor is

$$R = r_1 + r_2 \qquad \ldots (3)$$

from (1) Substituting $r = R_1 - r_2$ from eqn. (1) in equation (2) we have

$$R_2 = r_2 + \frac{(R_1 - r_2)\, r_1}{(R_1 - r_2) + r_1}$$

$$r_2 = R_2 - \sqrt{(R_1 - R_2)(R_1 - R_2)} \qquad \ldots (4)$$

If the total length of the cable is L meters, the length of the cable between far end and fault is L_1 meters, length of cable between test end and fault is L_2 meters and cross-section of conductor is uniform then

$$\frac{L_1}{L_2} = \frac{r_1}{r_2}$$ Add 1 to both sides

and $$\frac{L_1 + L_2}{L_2} = \frac{r_1 + r_2}{r_2}$$; but $L_1 + L_2 = L$

or then $$\frac{L}{L_2} = \frac{r_1 + r_2}{r_2}$$

$$L_2 = L \times \frac{r_2}{r_1 + r_2} \quad \ldots (5)$$

Thus the distance of fault from test end can be determined.

~~MURRAY~~ LOOP TESTS

These tests are performed for the location of either an earth fault or a short-circuit fault in underground cables provided that a sound cable runs along with the grounded or a short-circuited cable. In these ~~tests~~ the resistance of fault does not affect the results obtained except when the resistance of fault is very high. There are two loop tests usually used and are known as Murray loop and Varley loop tests. These tests employed the principle of Wheatstone bridge.

1. MURRAY LOOP TEST

The connection diagrams to locate earth fault and short-circuit fault by Murray loop test method are shown in Figures 2 and 3 respectively. Wheatstone bridge principle is used in these tests.

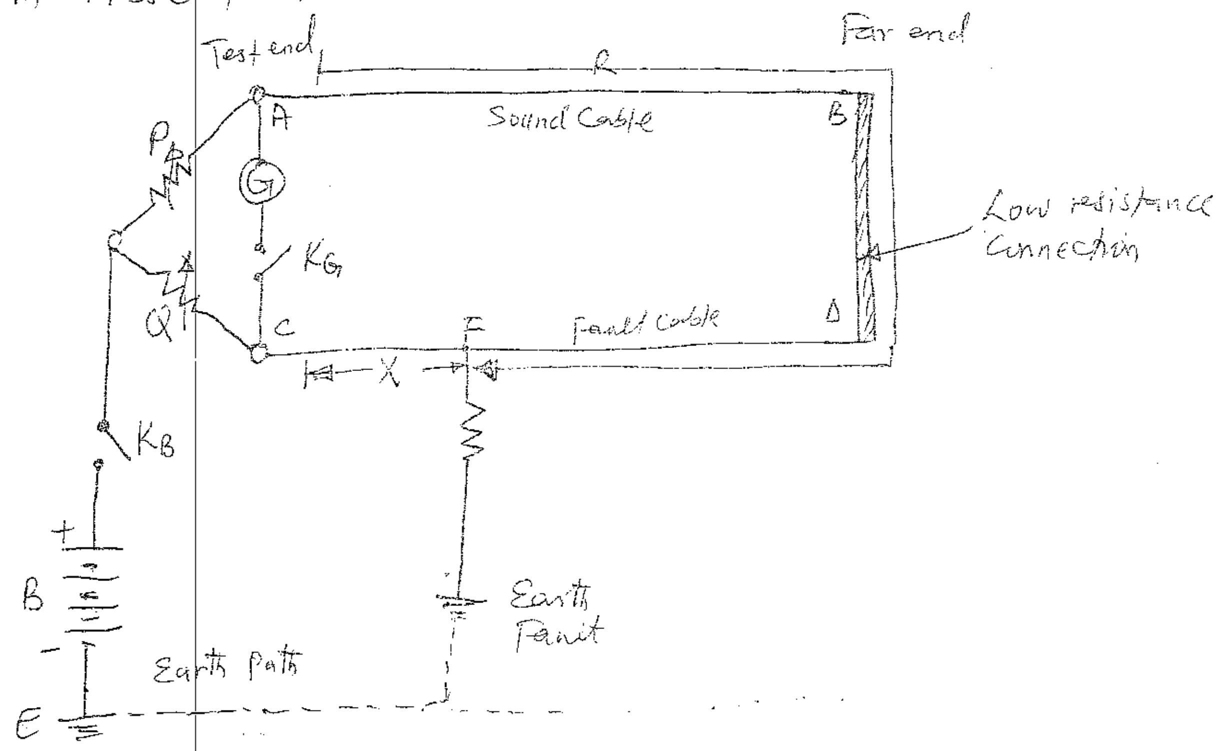


Figure 2: Murray Loop test (Earth fault)

AB is the sound cable and CD is the faulty cable; the earth fault occurring at point F

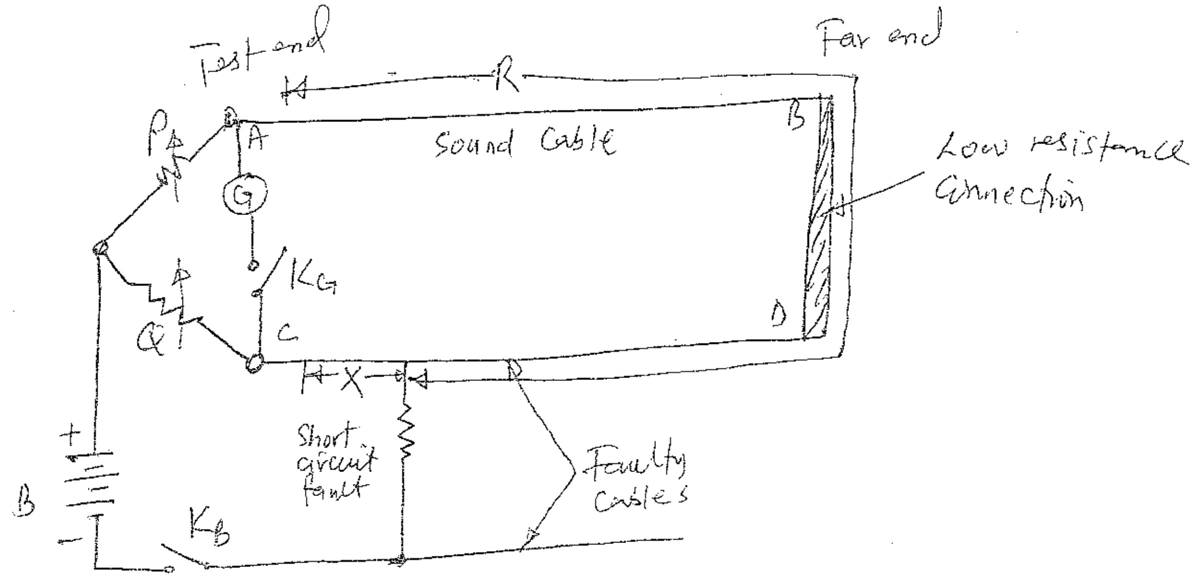


Figure 3: Murray loop test (short circuit fault).

P and Q are the two ratio arms consisting of step (variable) resistors and slide wire, G is the galvanometer, B is the battery, K_B is a battery key and K_G is a galvanometer key. The fault cable is connected to a second cable through low resistance link at the far end. Bridge is balanced by adjusting the resistance of ratio arms P and Q until the galvanometer indicates zero. Let R = resistance of the conductor loop upto the fault from the test end. while X = resistance of the other length of the loop.

In balanced condition of bridge

$$\frac{P}{Q} = \frac{R}{X}$$

Add 1 to both sides

$$\text{or} \quad \frac{P}{Q} + 1 = \frac{R}{X} + 1$$

If r is the resistance of each cable, then $R + X = 2r$

$$\text{or} \quad \frac{P+Q}{Q} = \frac{R+X}{X} = \frac{2r}{X}$$

Where 2r = twice resistance of one of the cable.

$$\text{or} \quad X = \frac{Q}{P+Q} \times 2r$$

Where r is resistance of one ~~alot~~ of the cables when free from fault. If l is the length of each cable in meters, then

resistance per meter length of cable $= r/l$

[illegible]

∴ Distance of fault from testing end $= \dfrac{\text{Resistance of the length of loop}}{\text{Resistance per meter length of cable}}$

$$d = \frac{X}{r/l} = \frac{lX}{r} = \frac{Q}{P+Q} \times 2r \cdot \frac{l}{r}$$

$$= \frac{Q}{P+Q} \times 2l \text{ meters} \qquad \ldots (6)$$

or $= \frac{Q}{P+Q} \times$ (loop length) meters.

The connection for short-circuit test in figure 3 are practically the same as in earth fault test except, that a portion of one of the short-circuiting cables is used instead of earth as return path in the battery circuit. Balance is obtained as before.

At balance $\dfrac{P}{Q} = \dfrac{R}{X}$; Add 1 to both sides gives $\dfrac{P+Q}{Q} = \dfrac{R+X}{X}$

or $\dfrac{Q}{P+Q} = \dfrac{X}{R+X}$ (inverted)

$$\text{or } X = \frac{Q}{P+Q}(R+X) \qquad \ldots (7)$$

Where $(R+X)$ is the total resistance of the loop and is assumed to be known.

Since P, Q and the total resistance ~~is~~ of the loop is known so X and hence the distance of fault from known end of ratio arm Q can be determined.

VARLEY LOOP TEST

In this test also sound cable is required in addition to the existing cable. The circuit diagrams to determine the location of ground fault and short-circuit fault by method of Varley loop tests are shown in Figure 4 and Figure 5 respectively.

The difference between Murray loop and Varley loop tests is that in Varley loop test provision is made for the measurement of total loop resistance instead of obtaining it from the relation $R = \rho \frac{l}{a}$.

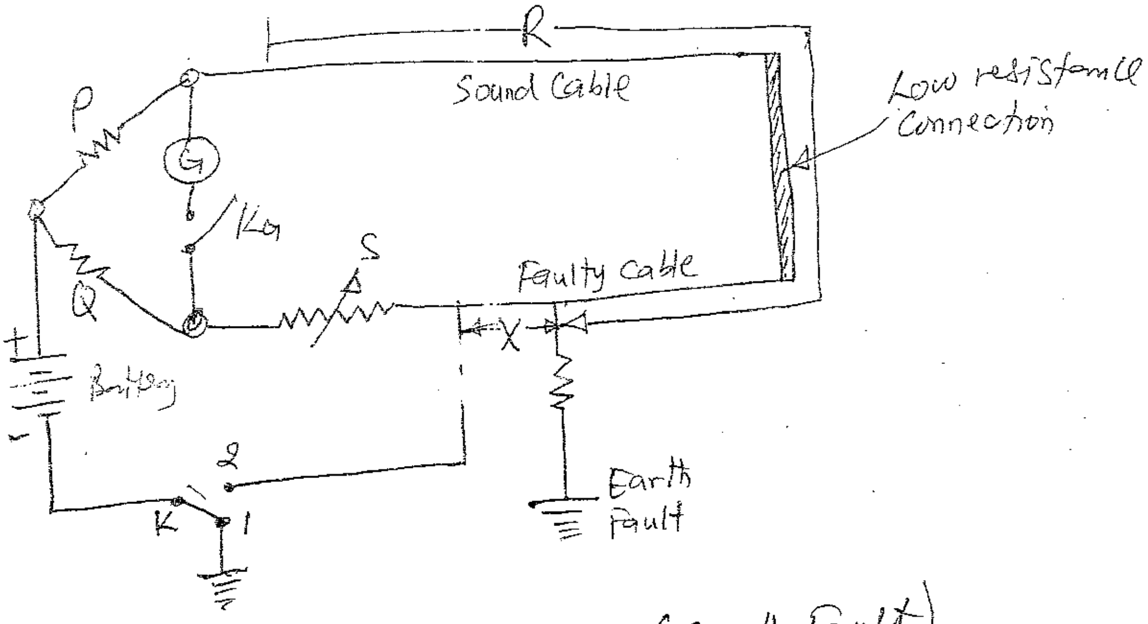


Figure 4: Varley loop test (Earth Fault)

In this test the ratio arms P and Q are fixed and balance is obtained by varying the known variable resistance S. Balance is obtained once the with ~~the~~ key in the battery circuit on stud 1 and then on stud 2.

At balance for earth fault or short-circuit fault with key K on stud 1.

$$\frac{P}{Q} = \frac{R}{X+S_1}$$

Where S_1 is the setting of Variable resistance S.

or $$\frac{P+Q}{Q} = \frac{R+X+S_1}{X+S_1}$$ Add 1 to both sides

or $$X = \frac{Q(R+X)-PS_1}{P+Q} \quad \ldots(8)$$

At balance for earth fault or Short-circuit fault when key K on stud 2,

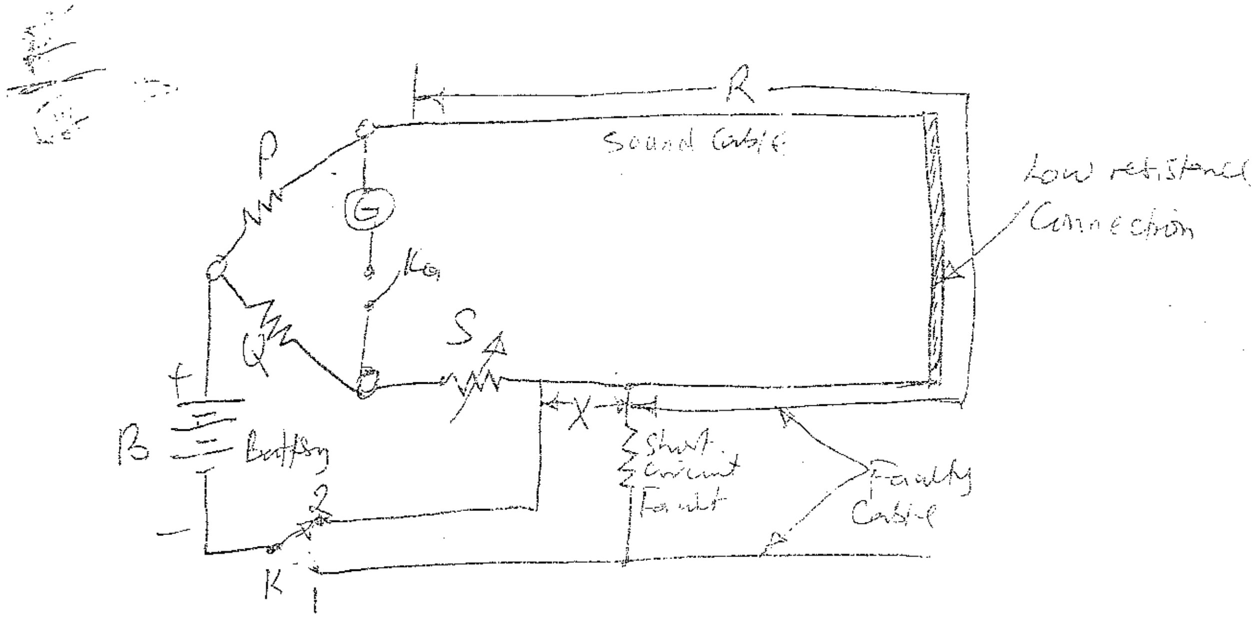


Figure 5: Varley Loop test (Short-circuit)

$$\frac{P}{Q} = \frac{R+X}{S_2}$$

Where S_2 is the new setting of Variable resistance S.

or $(R+X)Q = PS_2 \quad \ldots(9)$

or loop resistance $(R+X) = \frac{PS_2}{}$

From equations (8) and (9) we have

$$X = \frac{P(S_2 - S_1)}{P + Q} \quad \cdots (10)$$

Thus X can be determined from the known values of P, Q, S_2 and S_1.

Example 1.

In locating a ground fault by Varley loop test the following observations were recorded:
(i) With key K at stud 1, a balance was obtained when $P = 10$, $Q = 80$ and $S = 2{,}400\ \Omega$.
(ii) With key K at stud 2, a balance was obtained when $P = 10$, $Q = 80$ and $S = 3{,}400\ \Omega$.
Determine the distance of the fault from the testing end. Each conductor is 5 km long.

Soln: –

$P = 10\ \Omega$; $Q = 80\ \Omega$; $S_2 = 3{,}400\ \Omega$; $S_1 = 2400\ \Omega$

$$\therefore X = \frac{P(S_2 - S_1)}{P + Q} = \frac{10(3{,}400 - 2{,}400)}{10 + 80} = 111.11\ \Omega$$

$$\text{Loop resistance, } R + X = \frac{P}{Q} S_2 = \frac{10}{80} \times 3{,}400$$

$$\text{or } r = 425\ \Omega$$

$$\text{Resistance per km length of cable} = \frac{r}{l} = \frac{425}{10} = 42.5\ \Omega$$

$\because l = 2 \times 5$ i.e. twice loop length $= 10$

$$\text{Distance of fault from testing end} = \frac{X}{r/l} = \frac{111.11}{42.5} = 2.623 \text{ km}$$

=

Example 2

In a test by Murray loop for ground fault on 1,000 metres of cable having a resistance of 1.6 Ω per km, the faulty cable is looped with a sound cable of the same length and cross section. If the ratio of the other two arms of testing network at balance are in the ratio of 3 : 1, calculate the distance of the fault from the testing end of the cable.

Soln: –

The ratio of two arms, $\frac{P}{Q} = 3$

or $\frac{P+Q}{Q} = 3 + 1 = 4$

[from add. on both sides
$\frac{P}{Q} + 1 = \frac{3}{1} + 1$
$\frac{P+Q}{Q} = 3 + 1$]

Total resistance of loop is $(R+X)$

$\therefore R + X = \frac{1.6}{1000} \times 2 \times 1000$

$= 3.2\ \Omega$

From eqn (7) $X = \frac{Q}{P+Q}(R+X) = \frac{1}{4} \times 3.2 = 0.8\ \Omega$

Distance of fault from testing end is given by

$\frac{X}{r/l} = \frac{0.8}{1.6/1000} = 500$ metres

Transducer is an electronic device that converts energy from one form to another.